



Importance measures for two semi-coherent systems with shared components

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ABSTRACT

Importance measures are widely used for single components or pairs/groups of components in single reliability systems for evaluating their importance in the systems. In this paper, two semi-coherent systems with shared components are considered, and then the B-reliability importance and B-TDL (time-dependent-lifetime) importance measures are proposed for evaluating the importance of the shared components in both systems. Signature-based representations are presented next for facilitating the computation of the two proposed importance measures. The results established here are finally illustrated with some numerical examples involving consecutive- k systems with shared components. Applications of the results can be widely found for many practical systems in the field of reliability.

1. Introduction

Importance measures play a very useful role in the field of reliability for evaluating the relative importance of components in a system, as they facilitate the optimal design of the system and reasonable allocation of resources [1]. Importance measures are useful in many other fields as well such as bioinformatics, transportation, cloud computing, airline network, software development, management science, and risk analysis [2]. Recently, considerable work has been carried out in this direction; for example, Li et al. [3] proposed a method based on multi-valued decision diagrams and logical partial derivative for importance analysis of underframe connection system for the pantograph lower arm rod, and Fu et al. [4] introduced importance measures based methods for component reassignment problem of degrading components under hot and cold degradation termination, Huang et al. [5] presented a method based on survival signature for analyzing reliability importance of phased mission systems with different types of components. One may also refer to [1,6,7] for more related discussions in this regard.

The concept of importance measures was first proposed by Birnbaum [8] in 1969. Since then, many different types of importance measures have been proposed and studied, and all these can be categorized into three main classes: (1) Structure importance measures that evaluate the importance of positions in a system that components occupy, such as B-i.i.d. importance, B-structure importance, FV structure importance and BP structure importance; (2) Reliability importance measures that evaluate the importance of components to the reliability of a system, such as B-reliability importance, criticality reliability importance, Bayesian reliability importance and FV reliability importance;

and (3) Lifetime importance measures that evaluate the importance of components to the lifetime of a system, such as time-dependent-lifetime importance (TDL) measures like B-TDL importance, criticality TDL importance, FV TDL importance and BP TDL importance, and time-independent-lifetime (TIL) importance measures like BP TIL importance, $L_1 - L_6$ TIL importance and yield TIL importance. In fact, structure importance measures depend only on system structure, reliability importance measures depend on both system structure and component reliability, while lifetime importance measures depend on both system structure and component lifetime distribution. More detailed discussions on all these measures can be found in [1,2,9–11].

As one of the most basic importance measures, the Birnbaum reliability importance measure [8] of component i for a coherent system with structure function $\phi(X)$ is defined as $I_{Bs}(i; p) = I_{Bs}(i; p) + I_{Bf}(i; p)$, where $I_{Bs}(i; p) = \mathbb{P}\{\phi(X) = 1 | X_i = 1\} - \mathbb{P}\{\phi(X) = 1\}$ is the increase in system working probability if component i is given to be working and $I_{Bf}(i; p) = \mathbb{P}\{\phi(X) = 0 | X_i = 0\} - \mathbb{P}\{\phi(X) = 0\}$ is the increase in system failure probability if component i is given to be failed. The term $I_{Bs}(i; p)$ in Birnbaum reliability importance measure is also known as improvement potential importance or risk achievement importance. Note that reliability importance measures are essentially time-dependent, and they can be transformed to corresponding TDL importance measures by replacing component reliability p_i ($i = 1, \dots, n$) in $p = (p_1, \dots, p_n)$ with component reliability function $\bar{F}_i(t)$ ($i = 1, \dots, n$); see [10] for pertinent details.

Importance measures are not only useful in evaluating the importance of single components in a system, but they can also be applied for

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evaluating the importance of pairs/groups of components in a system. For example, for two independent components i and j in a system, Hong and Lie [12] defined their joint reliability importance and joint failure importance measures as

$$I_{JR}(i, j; p) = \frac{\partial^2 R(p)}{\partial p_i \partial p_j}, \quad I_{JF}(i, j; p) = \frac{\partial^2 [1 - R(p)]}{\partial q_i \partial q_j} = -I_{JR}(i, j; p),$$

respectively, where $R(p) = R(p_1, \dots, p_n)$ is the system reliability function with p_i being the reliability of component i and $q_i = 1 - p_i$ ($i = 1, \dots, n$). For any independent component pair i and j , their joint reliability importance is such that $-1 \leq I_{JR}(i, j; p) \leq 1$ and they are called reliability complements/substitutes if their joint reliability importance is nonnegative/nonpositive [13]. Joint reliability importance measure evaluates the interaction of components instead of their overall importance in the system [14]. For the latter one, Zio and Podofillini [15] proposed a second-order differential importance measure (DIM) for evaluating both the individual effects and the interactions of pairs of components.

In reliability applications, there exist a lot of practical systems that share components; for example, individual computers sharing the same server or gateway [16,17]. However, no importance measure has been proposed for evaluating the importance of shared components in the systems that share them. Existing research on systems with shared components focus mainly on their reliability or signature measures. Reliability of consecutive k -type systems (a type of system whose failure criteria depend on the number of consecutive working/failed components) in a zigzag/polygon structure with shared components has been derived by finite Markov chain imbedding approach (FMCIA) [18–21]. Joint reliability, instead of reliability, of consecutive k -type systems in a zigzag structure with shared components has also been derived by FMCIA in a new way [22–24]. Signature measures like joint signature, joint reliability signature, joint minimal/maximal signature, generalized joint signature, joint survival signature and multi-state joint signature have all been developed for describing the structural properties of different types of systems with shared components [16, 25–28].

In this work, we consider two semi-coherent systems with shared components and propose the B-reliability importance and B-TDL importance measures for evaluating the importance of these shared components in both systems. For single exchangeable components in single systems, their importance measures can be represented as a function of system signature and component lifetime distribution, which is similar to the well-known signature-based reliability function representation; see Eryilmaz [29] and Pitzen and Burkschat [30], for example. In this work, related discussions are generalized to the B-reliability importance and B-TDL importance measures for semi-coherent systems with shared independent and identically distributed (*i.i.d.*) components. Importance measures of components in consecutive k -type systems have attracted attention of researchers [31–33], and the results in this work have also been applied to two consecutive- k systems with shared components for the purpose of illustration. The theoretical results established here can also be applied to many different types of practical systems with shared components; for example, importance of nodes/edges shared by different paths in various types of networks (transportation networks, cancer gene networks and many others) can be compared. Applications of these results can also be found in many practical reliability systems such as smart street light systems, start-up tests and sensing systems. Main contributions of this paper can be summarized as follows: (1) B-reliability importance and B-TDL importance measures for shared components are proposed; (2) Signature-based representations are presented for the proposed importance measures; (3) The established results are applied to consecutive- k systems with shared components for illustrative purpose.

The rest of this paper proceeds as follows. In Sections 2 and 3, the B-reliability importance and B-TDL importance measures are proposed and studied for components shared by two semi-coherent systems for

evaluating their importance in both systems. In Section 4, signature-based representations are presented for the computation of the B-reliability importance and B-TDL importance measures of the *i.i.d.* components shared by two semi-coherent systems. Some numerical examples of consecutive- k systems with shared components are provided in Section 5 for illustrating the theoretical results developed in the preceding sections. Finally, some concluding remarks are made in Section 6.

2. B-reliability importance measure for two systems with shared components

Following the definition of B-reliability importance measure in Birnbaum [8] for components in a single system, we can propose B-reliability importance measure for components shared by two semi-coherent systems in an analogous way (by considering the increase in system working/failure probability when component i is given to be working/failed). Note that semi-coherent systems are monotone systems whose reliabilities will not decrease with an improvement in any component, which will simply be coherent systems if there is no irrelevant components in them. Components shared by two systems can be relevant to only one of the system or to both of them, which makes the two systems semi-coherent instead of coherent.

Consider two semi-coherent systems ϕ_1, ϕ_2 with n shared components, and denote the component state vector of the n components by $\mathbf{X} = (X_1, \dots, X_n)$, where $X_i = 1$ and $X_i = 0$ stand for the working state and failure state of component i , respectively. Let $p = (p_1, \dots, p_n)$, where $p_i = \mathbb{P}\{X_i = 1\}$, be the vector of probabilities of the components being in the working state. Let $q = (q_1, \dots, q_n)$, where $q_i = \mathbb{P}\{X_i = 0\}$, be the vector of probabilities of the components being in the failure state. Denote the structure functions of systems ϕ_1, ϕ_2 by $\phi_1(\mathbf{X}), \phi_2(\mathbf{X})$, respectively, and denote the event $\phi_1(\mathbf{X}) = a, \phi_2(\mathbf{X}) = b$, with $a, b \in \{0, 1\}$, by $A_{a,b}$. Then, for $a, b, c \in \{0, 1\}$, let $I_c^{ab}(i; p) = \mathbb{P}\{A_{a,b} | X_i = c\} - \mathbb{P}\{A_{a,b}\}$ be the increase in the probability that system ϕ_1 is in state a and system ϕ_2 is in state b when component i is given to be in state c . Let us denote $\phi_1(\mathbf{X}) = 1 - \phi_1(\mathbf{X}), \phi_2(\mathbf{X}) = 1 - \phi_2(\mathbf{X})$ and $(c_i, \mathbf{X}) = (X_1, \dots, X_{i-1}, c, X_{i+1}, \dots, X_n)$ (a vector generated by replacing the i th element in vector $\mathbf{X}$ by $c \in \{0, 1\}$).

Now, let us consider the formulas of $I_c^{ab}(i; p)$ given that component i is in the working state, i.e., $X_i = 1$ (or $c = 1$). For $a = b = 1$, along the lines of Kuo and Zhu [2], we find

$$\begin{aligned} I_1^{11}(i; p) &= \mathbb{P}\{A_{1,1} | X_i = 1\} - \mathbb{P}\{A_{1,1}\} \\ &= \mathbb{P}\{A_{1,1}, X_i = 1\} / \mathbb{P}\{X_i = 1\} - \mathbb{P}\{A_{1,1}\} \\ &= p_i^{-1} \mathbb{E}[X_i \phi_1(\mathbf{X}) \phi_2(\mathbf{X})] - \mathbb{E}[\phi_1(\mathbf{X}) \phi_2(\mathbf{X})] \\ &= p_i^{-1} [\mathbb{E}(X_i) \mathbb{E}[\phi_1(\mathbf{X}) \phi_2(\mathbf{X})] + \text{Cov}[X_i, \phi_1(\mathbf{X}) \phi_2(\mathbf{X})]] \\ &\quad - \mathbb{E}[\phi_1(\mathbf{X}) \phi_2(\mathbf{X})] \\ &= p_i^{-1} \text{Cov}[X_i, \phi_1(\mathbf{X}) \phi_2(\mathbf{X})], \end{aligned}$$

where from the decomposition $\phi_1(\mathbf{X}) \phi_2(\mathbf{X}) = X_i \phi_1(1_i, \mathbf{X}) \phi_2(1_i, \mathbf{X}) + (1 - X_i) \phi_1(0_i, \mathbf{X}) \phi_2(0_i, \mathbf{X})$, we have

$$\begin{aligned} \text{Cov}[X_i, \phi_1(\mathbf{X}) \phi_2(\mathbf{X})] &= \text{Cov}\{X_i, X_i [\phi_1(1_i, \mathbf{X}) \phi_2(1_i, \mathbf{X}) - \phi_1(0_i, \mathbf{X}) \phi_2(0_i, \mathbf{X})]\} \\ &= \mathbb{E}\{X_i^2 [\phi_1(1_i, \mathbf{X}) \phi_2(1_i, \mathbf{X}) - \phi_1(0_i, \mathbf{X}) \phi_2(0_i, \mathbf{X})]\} \\ &\quad - \mathbb{E}(X_i) \mathbb{E}\{X_i [\phi_1(1_i, \mathbf{X}) \phi_2(1_i, \mathbf{X}) - \phi_1(0_i, \mathbf{X}) \phi_2(0_i, \mathbf{X})]\} \\ &= [1 - \mathbb{E}(X_i)] \mathbb{E}(X_i) \mathbb{E}[\phi_1(1_i, \mathbf{X}) \phi_2(1_i, \mathbf{X}) - \phi_1(0_i, \mathbf{X}) \phi_2(0_i, \mathbf{X})] \\ &= p_i q_i \mathbb{E}[\phi_1(1_i, \mathbf{X}) \phi_2(1_i, \mathbf{X}) - \phi_1(0_i, \mathbf{X}) \phi_2(0_i, \mathbf{X})], \end{aligned}$$

which means $I_1^{11}(i; p) = q_i \mathbb{E}[\phi_1(1_i, \mathbf{X}) \phi_2(1_i, \mathbf{X}) - \phi_1(0_i, \mathbf{X}) \phi_2(0_i, \mathbf{X})]$. Using similar arguments, we find the formulas for $I_1^{ab}(i; p)$, $a, b \in \{0, 1\}$, as follows:

$$\begin{aligned} I_1^{11}(i; p) &= q_i \mathbb{E}[\phi_1(1_i, \mathbf{X}) \phi_2(1_i, \mathbf{X}) - \phi_1(0_i, \mathbf{X}) \phi_2(0_i, \mathbf{X})], \\ I_1^{10}(i; p) &= q_i \mathbb{E}[\phi_1(1_i, \mathbf{X}) \phi_2(1_i, \mathbf{X}) - \phi_1(0_i, \mathbf{X}) \phi_2(0_i, \mathbf{X})], \end{aligned}$$

$$I_1^{01}(i; p) = q_i \mathbb{E}[\bar{\phi}_1(1_i, X) \phi_2(1_i, X) - \bar{\phi}_1(0_i, X) \phi_2(0_i, X)],$$

$$I_1^{00}(i; p) = q_i \mathbb{E}[\bar{\phi}_1(1_i, X) \bar{\phi}_2(1_i, X) - \bar{\phi}_1(0_i, X) \bar{\phi}_2(0_i, X)].$$

We can also consider the formulas of $I_c^{ab}(i; p)$ given that component i is in the failure state, namely, $X_i = 0$ (or $c = 0$). For $a = b = 1$, we find

$$\begin{aligned} I_0^{11}(i; p) &= \mathbb{P}\{A_{1,1} \mid X_i = 0\} - \mathbb{P}\{A_{1,1}\} \\ &= \mathbb{P}\{A_{1,1}, X_i = 0\} / \mathbb{P}\{X_i = 0\} - \mathbb{P}\{A_{1,1}\} \\ &= q_i^{-1} \mathbb{E}[(1 - X_i) \phi_1(X) \phi_2(X)] - \mathbb{E}[\phi_1(X) \phi_2(X)] \\ &= q_i^{-1} \{\mathbb{E}[(1 - X_i) \mathbb{E}[\phi_1(X) \phi_2(X)] + Cov[1 - X_i, \phi_1(X) \phi_2(X)]] \\ &\quad - \mathbb{E}[\phi_1(X) \phi_2(X)]\} \\ &= q_i^{-1} Cov[1 - X_i, \phi_1(X) \phi_2(X)], \end{aligned}$$

where

$$\begin{aligned} Cov[1 - X_i, \phi_1(X) \phi_2(X)] &= -Cov[X_i, \phi_1(X) \phi_2(X)] \\ &= -p_i q_i \mathbb{E}[\phi_1(1_i, X) \phi_2(1_i, X) - \phi_1(0_i, X) \phi_2(0_i, X)], \end{aligned}$$

which means $I_0^{11}(i; p) = p_i \mathbb{E}[\phi_1(0_i, X) \phi_2(0_i, X) - \phi_1(1_i, X) \phi_2(1_i, X)]$. Using similar arguments, we have the formulas for $I_0^{ab}(i; p)$, $a, b \in \{0, 1\}$, as follows:

$$\begin{aligned} I_0^{11}(i; p) &= p_i \mathbb{E}[\phi_1(0_i, X) \phi_2(0_i, X) - \phi_1(1_i, X) \phi_2(1_i, X)], \\ I_0^{10}(i; p) &= p_i \mathbb{E}[\phi_1(0_i, X) \bar{\phi}_2(0_i, X) - \phi_1(1_i, X) \bar{\phi}_2(1_i, X)], \\ I_0^{01}(i; p) &= p_i \mathbb{E}[\bar{\phi}_1(0_i, X) \phi_2(0_i, X) - \bar{\phi}_1(1_i, X) \phi_2(1_i, X)], \\ I_0^{00}(i; p) &= p_i \mathbb{E}[\bar{\phi}_1(0_i, X) \bar{\phi}_2(0_i, X) - \bar{\phi}_1(1_i, X) \bar{\phi}_2(1_i, X)]. \end{aligned}$$

With the formulas of $I_c^{ab}(i; p)$, $a, b, c \in \{0, 1\}$, presented above, we can further discuss their signs and define B-reliability importance measure of component i by summing the positive ones among them. First of all, we notice that $p_i I_1^{ab}(i; p) = -q_i I_0^{ab}(i; p)$ for all $a, b \in \{0, 1\}$, which means $I_1^{ab}(i; p)$ and $I_0^{ab}(i; p)$ have different signs. Then, from the formulas of $I_c^{ab}(i; p)$, we find that

$$\begin{aligned} I_1^{11}(i; p) &= q_i \mathbb{E}[\phi_1(1_i, X) \phi_2(1_i, X) - \phi_1(0_i, X) \phi_2(0_i, X)] \\ &= q_i \mathbb{E}[\phi_{(ser)}(1_i, X) - \phi_{(ser)}(0_i, X)], \\ I_0^{00}(i; p) &= p_i \mathbb{E}[\bar{\phi}_1(0_i, X) \bar{\phi}_2(0_i, X) - \bar{\phi}_1(1_i, X) \bar{\phi}_2(1_i, X)] \\ &= p_i \mathbb{E}[\phi_{(par)}(1_i, X) - \phi_{(par)}(0_i, X)], \end{aligned}$$

where system $\phi_{(ser)}/\phi_{(par)}$ is a series/parallel connection of semi-coherent systems ϕ_1 and ϕ_2 . The B-reliability importance measure of component i in system $\phi_{(ser)}/\phi_{(par)}$ must be nonnegative, namely, $I_1^{11}(i; p) \geq 0$, $I_0^{00}(i; p) \geq 0$, which implies that $I_0^{11}(i; p) \leq 0$, $I_1^{00}(i; p) \leq 0$. Besides that, the signs of $I_1^{10}(i; p)$, $I_0^{10}(i; p)$, $I_1^{01}(i; p)$, $I_0^{01}(i; p)$ are not fixed. For example,

$$\begin{aligned} I_1^{10}(i; p) &= q_i \mathbb{E}[\phi_1(1_i, X) \bar{\phi}_2(1_i, X) - \phi_1(0_i, X) \bar{\phi}_2(0_i, X)] \\ &= q_i \{\mathbb{E}[\phi_1(1_i, X) - \phi_1(0_i, X)] - \mathbb{E}[\phi_{(ser)}(1_i, X) - \phi_{(ser)}(0_i, X)]\}, \end{aligned}$$

where $\mathbb{E}[\phi_1(1_i, X) - \phi_1(0_i, X)]$ is the B-reliability importance measure of component i in system ϕ_1 . Consider a special case of systems ϕ_1, ϕ_2 having reliability functions $R_1(p) = f_1(p)f_{12}(p)$, $R_2(p) = f_2(p)f_{12}(p)$ and their series connection $\phi_{(ser)}$ having the reliability function $R_{(ser)}(p) = f_1(p)f_2(p) \cdot f_{12}(p)$, where $f_1(p), f_2(p), f_{12}(p)$ are reliability functions of three independent modules with components only in system ϕ_1 , components only in system ϕ_2 , and components in both systems ϕ_1 and ϕ_2 , respectively. Then, we have $I_1^{10}(i; p) = q_i \left[\frac{\partial R_1(p)}{\partial p_i} - \frac{\partial R_{(ser)}(p)}{\partial p_i} \right]$, which means when component i is only in system ϕ_1 , we have

$$I_1^{10}(i; p) = q_i \frac{\partial f_1(p)}{\partial p_i} [1 - f_2(p)] f_{12}(p) \geq 0;$$

when component i is only in system ϕ_2 , we have

$$I_1^{10}(i; p) = -q_i f_1(p) \frac{\partial f_2(p)}{\partial p_i} f_{12}(p) \leq 0;$$

and when component i is in both systems ϕ_1 and ϕ_2 , we have

$$I_1^{10}(i; p) = q_i f_1(p) [1 - f_2(p)] \frac{\partial f_{12}(p)}{\partial p_i} \geq 0.$$

Based on the above discussions, we can define the B-reliability importance measure for two semi-coherent systems with shared components as follows.

Definition 2.1. The B-reliability importance measure of component i in semi-coherent systems ϕ_1 and ϕ_2 is defined as

$$\begin{aligned} I(i; p) &= I_1^{11}(i; p) + I_0^{00}(i; p) \\ &= q_i \mathbb{E}[\phi_{(ser)}(1_i, X) - \phi_{(ser)}(0_i, X)] + p_i \mathbb{E}[\phi_{(par)}(1_i, X) - \phi_{(par)}(0_i, X)], \end{aligned}$$

where $\mathbb{E}[\phi_{(ser)}(1_i, X) - \phi_{(ser)}(0_i, X)]$ is the B-reliability importance measure of component i in the series connection of systems ϕ_1 and ϕ_2 , and $\mathbb{E}[\phi_{(par)}(1_i, X) - \phi_{(par)}(0_i, X)]$ is the B-reliability importance measure of component i in the parallel connection of systems ϕ_1 and ϕ_2 .

Remark 2.1. Only $I_1^{11}(i; p)$, $I_0^{00}(i; p)$ are contained in the definition not only because they are always positive, but also because $I_1^{11}(i; p)$ is the increase in the working probability for both systems if component i is given to be working and $I_0^{00}(i; p)$ is the increase in the failure probability for both systems if component i is given to be failed. Besides, even if $I_1^{10}(i; p)$, $I_0^{01}(i; p)$ are not contained in the definition, their signs and values are useful to show the unbalanced importance of component i to systems ϕ_1 and ϕ_2 , which means it will also be a possible choice to define $I(i; p)$ as a vector of them or a linear combination of their absolute values from a practical viewpoint. No matter which way is preferred, $I_1^{11}(i; p)$, $I_0^{00}(i; p)$ will always be the most important part of the measure. See [Example 5.1](#) and [Appendix A.1](#) for some related numerical calculations and associated explanations.

The above definition can be generalized to the case of more than two semi-coherent systems with shared components readily as follows.

Definition 2.2. The B-reliability importance measure of component i in semi-coherent systems $\phi_1, \dots, \phi_l$ is defined as

$$\begin{aligned} I(i; p) &= I_1^1(i; p) + I_0^0(i; p) \\ &= q_i \mathbb{E}[\phi_{(ser)}(1_i, X) - \phi_{(ser)}(0_i, X)] + p_i \mathbb{E}[\phi_{(par)}(1_i, X) - \phi_{(par)}(0_i, X)], \end{aligned}$$

where $\mathbb{E}[\phi_{(ser)}(1_i, X) - \phi_{(ser)}(0_i, X)]$ is the B-reliability importance measure of component i in the series connection of systems $\phi_1, \dots, \phi_l$, and $\mathbb{E}[\phi_{(par)}(1_i, X) - \phi_{(par)}(0_i, X)]$ is the B-reliability importance measure of component i in the parallel connection of systems $\phi_1, \dots, \phi_l$.

Remark 2.2. The B-reliability importance measure of component i can also be given as

$$I(i; p) = q_i [R_{(ser)}(1_i, p) - R_{(ser)}(0_i, p)] + p_i [R_{(par)}(1_i, p) - R_{(par)}(0_i, p)],$$

where $R_{(ser)}(p)$ is the reliability function of the series connection of systems $\phi_1, \dots, \phi_l$ and $R_{(par)}(p)$ is the reliability function of the parallel connection of systems $\phi_1, \dots, \phi_l$.

3. B-TDL importance measure for two systems with shared components

For the two semi-coherent systems ϕ_1, ϕ_2 with n shared components considered earlier in Section 2, their Birnbaum time-dependent-lifetime importance (B-TDL importance) measure can also be proposed by considering the increase in system working/failure probability when component i is given to be working/failed at specific times. Denote the component state vector at time t of the n components by $X(t) = (X_1(t), \dots, X_n(t))$, where $X_i(t) = 1$ and $X_i(t) = 0$ stand for the working state and failure state of component i at time t , respectively. Let $F = (F_1, \dots, F_n)$, where F_i is the cumulative lifetime distribution function of component i , and $\bar{F}_i = 1 - F_i$. Denote the structure functions of systems ϕ_1, ϕ_2 by $\phi_1(X), \phi_2(X)$, respectively, and the event $\phi_1[X(t_1)] =$

$a, \phi_2[X(t_2)] = b$, with $a, b \in \{0, 1\}$, by $A_{a,b}$. Then, for $a, b, c, d \in \{0, 1\}$, let $I_{cd}^{ab}(i; F) = \mathbb{P}\{A_{a,b} | X_i(t_1) = c, X_i(t_2) = d\} - \mathbb{P}\{A_{a,b}\}$ be the increase in the probability that system ϕ_1 is in state a at time t_1 and system ϕ_2 is in state b at time t_2 when component i is given to be in state c at time t_1 and in state d at time t_2 . Let us also denote $\bar{\phi}_1(X) = 1 - \phi_1(X)$, $\bar{\phi}_2(X) = 1 - \phi_2(X)$ and $(c_i, X(t)) = (X_1(t), \dots, X_{i-1}(t), c, X_{i+1}(t), \dots, X_n(t))$ (a vector generated by replacing the i th element in vector $X(t)$ by $c \in \{0, 1\}$). Then, further discussions on the formulas for $I_{cd}^{ab}(i; F)$ are provided below. Without loss of generality, we only consider the case $t_1 \leq t_2$ in these discussions, which means system ϕ_1 is observed and used before system ϕ_2 .

Now, let us consider the formulas of $I_{cd}^{ab}(i; F)$ given that component i is in the working state at time both times t_1 and t_2 , i.e., $X_i(t_1) = X_i(t_2) = 1$ (or $c = d = 1$). For $a = b = 1$, we find

$$\begin{aligned} I_{11}^{11}(i; F) &= \mathbb{P}\{A_{1,1} | X_i(t_1) = 1, X_i(t_2) = 1\} - \mathbb{P}\{A_{1,1}\} \\ &= \mathbb{P}\{A_{1,1}, X_i(t_2) = 1\} / \mathbb{P}\{X_i(t_2) = 1\} - \mathbb{P}\{A_{1,1}\} \\ &= [\bar{F}_i(t_2)]^{-1} \mathbb{E}\{X_i(t_2)\phi_1[X(t_1)]\phi_2[X(t_2)]\} \\ &\quad - \mathbb{E}\{\phi_1[X(t_1)]\phi_2[X(t_2)]\} \\ &= [\bar{F}_i(t_2)]^{-1} \mathbb{E}[X_i(t_2)]\mathbb{E}\{\phi_1[X(t_1)]\phi_2[X(t_2)]\} \\ &\quad + [\bar{F}_i(t_2)]^{-1} \text{Cov}\{X_i(t_2), \phi_1[X(t_1)]\phi_2[X(t_2)]\} \\ &\quad - \mathbb{E}\{\phi_1[X(t_1)]\phi_2[X(t_2)]\} \\ &= [\bar{F}_i(t_2)]^{-1} \text{Cov}\{X_i(t_2), \phi_1[X(t_1)]\phi_2[X(t_2)]\}, \end{aligned}$$

where from the decomposition

$$\begin{aligned} \phi_1[X(t_1)]\phi_2[X(t_2)] &= X_i(t_1)X_i(t_2)\phi_1[1_i, X(t_1)]\phi_2[1_i, X(t_2)] \\ &\quad + X_i(t_1)[1 - X_i(t_2)]\phi_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)] \\ &\quad + [1 - X_i(t_1)][1 - X_i(t_2)]\phi_1[0_i, X(t_1)]\phi_2[0_i, X(t_2)], \end{aligned}$$

we have

$$\begin{aligned} \text{Cov}\{X_i(t_2), \phi_1[X(t_1)]\phi_2[X(t_2)]\} &= \text{Cov}\{X_i(t_2), X_i(t_1)X_i(t_2)\phi_1[1_i, X(t_1)]\phi_2[1_i, X(t_2)] \\ &\quad + X_i(t_1)[1 - X_i(t_2)]\phi_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)] \\ &\quad + [1 - X_i(t_1)][1 - X_i(t_2)]\phi_1[0_i, X(t_1)]\phi_2[0_i, X(t_2)]\} \\ &= \{1 - \mathbb{E}[X_i(t_2)]\}\mathbb{E}\{X_i(t_1)X_i(t_2)\phi_1[1_i, X(t_1)]\phi_2[1_i, X(t_2)]\} \\ &\quad - \mathbb{E}[X_i(t_2)]\mathbb{E}\{X_i(t_1)[1 - X_i(t_2)]\phi_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)]\} \\ &\quad - \mathbb{E}[X_i(t_2)]\mathbb{E}\{[1 - X_i(t_1)][1 - X_i(t_2)]\phi_1[0_i, X(t_1)]\phi_2[0_i, X(t_2)]\} \\ &= F_i(t_2)\bar{F}_i(t_2)\mathbb{E}\{\phi_1[1_i, X(t_1)]\phi_2[1_i, X(t_2)]\} \\ &\quad - \bar{F}_i(t_2)[F_i(t_2) - F_i(t_1)]\mathbb{E}\{\phi_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)]\} \\ &\quad - \bar{F}_i(t_2)F_i(t_1)\mathbb{E}\{\phi_1[0_i, X(t_1)]\phi_2[0_i, X(t_2)]\}, \end{aligned}$$

which means that

$$\begin{aligned} I_{11}^{11}(i; F) &= F_i(t_2)\mathbb{E}\{\phi_1[1_i, X(t_1)]\phi_2[1_i, X(t_2)] - \phi_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)]\} \\ &\quad + F_i(t_1)\mathbb{E}\{\phi_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)] \\ &\quad - \phi_1[0_i, X(t_1)]\phi_2[0_i, X(t_2)]\}. \end{aligned}$$

Using similar arguments, we find the formulas for $I_{11}^{ab}(i; F)$, $a, b \in \{0, 1\}$ as follows:

$$\begin{aligned} I_{11}^{11}(i; F) &= F_i(t_2)\mathbb{E}\{\phi_1[1_i, X(t_1)]\phi_2[1_i, X(t_2)] - \phi_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)]\} \\ &\quad + F_i(t_1)\mathbb{E}\{\phi_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)] \\ &\quad - \phi_1[0_i, X(t_1)]\phi_2[0_i, X(t_2)]\}, \\ I_{11}^{10}(i; F) &= F_i(t_2)\mathbb{E}\{\phi_1[1_i, X(t_1)]\bar{\phi}_2[1_i, X(t_2)] - \phi_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)]\} \\ &\quad + F_i(t_1)\mathbb{E}\{\phi_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)] \\ &\quad - \phi_1[0_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)]\}, \\ I_{11}^{01}(i; F) &= F_i(t_2)\mathbb{E}\{\bar{\phi}_1[1_i, X(t_1)]\phi_2[1_i, X(t_2)] - \bar{\phi}_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)]\} \\ &\quad + F_i(t_1)\mathbb{E}\{\bar{\phi}_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)] \\ &\quad - \bar{\phi}_1[0_i, X(t_1)]\phi_2[0_i, X(t_2)]\}, \\ I_{11}^{00}(i; F) &= F_i(t_2)\mathbb{E}\{\bar{\phi}_1[1_i, X(t_1)]\bar{\phi}_2[1_i, X(t_2)] - \bar{\phi}_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)]\} \\ &\quad + F_i(t_1)\mathbb{E}\{\bar{\phi}_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)] \\ &\quad - \bar{\phi}_1[0_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)]\}. \end{aligned}$$

$$\begin{aligned} &+ F_i(t_1)\mathbb{E}\{\bar{\phi}_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)] \\ &- \bar{\phi}_1[0_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)]\}. \end{aligned}$$

Note that $I_{11}^{ab}(i; F)$, $a, b \in \{0, 1\}$, will reduce to $I_1^{ab}(i; p)$, $a, b \in \{0, 1\}$, in Section 2 if $t_1 = t_2 = t$.

We can also consider the formulas of $I_{cd}^{ab}(i; F)$ given that component i is in the working state at time t_1 and in the failure state at time t_2 , i.e., $X_i(t_1) = 1, X_i(t_2) = 0, t_1 < t_2$ (or $c = 1, d = 0$). Using similar arguments, we find the formulas for $I_{10}^{ab}(i; F)$, $a, b \in \{0, 1\}$, as follows:

$$\begin{aligned} I_{10}^{11}(i; F) &= \bar{F}_i(t_2)\mathbb{E}\{\phi_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)] - \phi_1[1_i, X(t_1)]\phi_2[1_i, X(t_2)]\} \\ &\quad + F_i(t_1)\mathbb{E}\{\phi_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)] \\ &\quad - \phi_1[0_i, X(t_1)]\phi_2[0_i, X(t_2)]\}, \\ I_{10}^{10}(i; F) &= \bar{F}_i(t_2)\mathbb{E}\{\phi_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)] - \phi_1[1_i, X(t_1)]\bar{\phi}_2[1_i, X(t_2)]\} \\ &\quad + F_i(t_1)\mathbb{E}\{\phi_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)] \\ &\quad - \phi_1[0_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)]\}, \\ I_{10}^{01}(i; F) &= \bar{F}_i(t_2)\mathbb{E}\{\bar{\phi}_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)] - \bar{\phi}_1[1_i, X(t_1)]\phi_2[1_i, X(t_2)]\} \\ &\quad + F_i(t_1)\mathbb{E}\{\bar{\phi}_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)] \\ &\quad - \bar{\phi}_1[0_i, X(t_1)]\phi_2[0_i, X(t_2)]\}, \\ I_{10}^{00}(i; F) &= \bar{F}_i(t_2)\mathbb{E}\{\bar{\phi}_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)] - \bar{\phi}_1[1_i, X(t_1)]\bar{\phi}_2[1_i, X(t_2)]\} \\ &\quad + F_i(t_1)\mathbb{E}\{\bar{\phi}_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)] \\ &\quad - \bar{\phi}_1[0_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)]\}. \end{aligned}$$

Similarly, consider the formulas of $I_{cd}^{ab}(i; F)$ given that component i is in the failure state at both times t_1 and t_2 , i.e., $X_i(t_1) = X_i(t_2) = 0$ (or $c = 1, d = 0$). We then find the formulas for $I_{00}^{ab}(i; F)$, $a, b \in \{0, 1\}$, as follows:

$$\begin{aligned} I_{00}^{11}(i; F) &= \bar{F}_i(t_2)\mathbb{E}\{\phi_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)] - \phi_1[1_i, X(t_1)]\phi_2[1_i, X(t_2)]\} \\ &\quad + \bar{F}_i(t_1)\mathbb{E}\{\phi_1[0_i, X(t_1)]\phi_2[0_i, X(t_2)] \\ &\quad - \phi_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)]\}, \\ I_{00}^{10}(i; F) &= \bar{F}_i(t_2)\mathbb{E}\{\phi_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)] - \phi_1[1_i, X(t_1)]\bar{\phi}_2[1_i, X(t_2)]\} \\ &\quad + \bar{F}_i(t_1)\mathbb{E}\{\phi_1[0_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)] \\ &\quad - \phi_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)]\}, \\ I_{00}^{01}(i; F) &= \bar{F}_i(t_2)\mathbb{E}\{\bar{\phi}_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)] - \bar{\phi}_1[1_i, X(t_1)]\phi_2[1_i, X(t_2)]\} \\ &\quad + \bar{F}_i(t_1)\mathbb{E}\{\bar{\phi}_1[0_i, X(t_1)]\phi_2[0_i, X(t_2)] \\ &\quad - \bar{\phi}_1[1_i, X(t_1)]\phi_2[0_i, X(t_2)]\}, \\ I_{00}^{00}(i; F) &= \bar{F}_i(t_2)\mathbb{E}\{\bar{\phi}_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)] - \bar{\phi}_1[1_i, X(t_1)]\bar{\phi}_2[1_i, X(t_2)]\} \\ &\quad + \bar{F}_i(t_1)\mathbb{E}\{\bar{\phi}_1[0_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)] \\ &\quad - \bar{\phi}_1[1_i, X(t_1)]\bar{\phi}_2[0_i, X(t_2)]\}. \end{aligned}$$

Note that $I_{00}^{ab}(i; F)$, $a, b \in \{0, 1\}$, will reduce to $I_0^{ab}(i; p)$, $a, b \in \{0, 1\}$, in Section 2 if $t_1 = t_2 = t$.

With the formulas for $I_{cd}^{ab}(i; F)$, $a, b, c, d \in \{0, 1\}$, presented above, we can further discuss their signs and define the B-TDL importance measure of component i by summing the positive ones among them. As in Section 2, we have $I_{11}^{11}(i; F) \geq 0$, $I_{11}^{00}(i; F) \leq 0$, $I_{10}^{10}(i; F) \geq 0$, $I_{10}^{01}(i; F) \leq 0$, $I_{00}^{11}(i; F) \leq 0$, $I_{00}^{00}(i; F) \geq 0$, and the signs of $I_{11}^{10}(i; F)$, $I_{11}^{01}(i; F)$, $I_{10}^{11}(i; F)$, $I_{10}^{00}(i; F)$, $I_{00}^{10}(i; F)$, $I_{00}^{01}(i; F)$ are not fixed. For example, for $t_1 = t_2 = t$, we have

$$I_{11}^{10}(i; F) = F_i(t)\mathbb{E}\{\phi_1[1_i, X(t)]\bar{\phi}_2[1_i, X(t)] - \phi_1[0_i, X(t)]\bar{\phi}_2[0_i, X(t)]\},$$

whose sign is not fixed as in Section 2.

Based on the above discussions, we can define the B-TDL importance measure for two semi-coherent systems with shared components as follows.

Definition 3.1. For $t_1 < t_2$, the B-TDL importance measure of component i for systems ϕ_1 and ϕ_2 is defined as

$$I(i; F) = I_{11}^{11}(i; F) + I_{10}^{10}(i; F) + I_{00}^{00}(i; F)$$

$$\begin{aligned}
&= F_i(t_1) \mathbb{E}\{\phi_1[1_i, \mathbf{X}(t_1)] - \phi_1[0_i, \mathbf{X}(t_1)]\} \\
&\quad + \bar{F}_i(t_2) \mathbb{E}\{\phi_2[1_i, \mathbf{X}(t_2)] - \phi_2[0_i, \mathbf{X}(t_2)]\} \\
&\quad + \bar{F}_i(t_1) \mathbb{E}\{\bar{\phi}_1[0_i, \mathbf{X}(t_1)]\bar{\phi}_2[0_i, \mathbf{X}(t_2)] - \bar{\phi}_1[1_i, \mathbf{X}(t_1)]\bar{\phi}_2[0_i, \mathbf{X}(t_2)]\} \\
&\quad + F_i(t_2) \mathbb{E}\{\phi_1[1_i, \mathbf{X}(t_1)]\phi_2[1_i, \mathbf{X}(t_2)] - \phi_1[1_i, \mathbf{X}(t_1)]\phi_2[0_i, \mathbf{X}(t_2)]\}.
\end{aligned}$$

Remark 3.1. In this definition, system ϕ_1 is observed at time t_1 and system ϕ_2 is observed at time t_2 , where t_1 is assumed to be less than t_2 without loss of generality since systems ϕ_1 and ϕ_2 can be interchanged if necessary. Only $I_{11}^{11}(i; F)$, $I_{10}^{10}(i; F)$, $I_{00}^{00}(i; F)$ are contained in the definition not only because they are always positive, but also because $I_{11}^{11}(i; F)$ is the increase in the working probability for system ϕ_1 at time t_1 and system ϕ_2 at time t_2 if component i is given to be working at both times t_1 and t_2 , $I_{10}^{10}(i; F)$ is the increase in probability for the working of system ϕ_1 at time t_1 and the failure of system ϕ_2 at time t_2 if component i is given to be working at time t_1 and failed at time t_2 , and $I_{00}^{00}(i; F)$ is the increase in the failure probability for system ϕ_1 at time t_1 and system ϕ_2 at time t_2 if component i is given to be failed at both times t_1 and t_2 . Besides, even if other terms are not contained in the definition, their signs and values are useful to show the unbalanced importance of component i to systems ϕ_1 at time t_1 and ϕ_2 at time t_2 , which means it will also be a possible choice to define $I(i; F)$ as a vector of them or a linear combination of their absolute values from a practical viewpoint. No matter which way is preferred, $I_{11}^{11}(i; F)$, $I_{10}^{10}(i; F)$, $I_{00}^{00}(i; F)$ will always be the most important part of the measure. See [Example 5.2](#) and [Appendix A.2](#) for some related numerical calculations and associated explanations.

Remark 3.2. The B-TDL importance measure of component i can also be given as

$$\begin{aligned}
I(i; F) &= F_i(t_1)[R_1(1_i, F; t_1) - R_1(0_i, F; t_1)] \\
&\quad + \bar{F}_i(t_2)[R_2(1_i, F; t_2) - R_2(0_i, F; t_2)] \\
&\quad + \bar{F}_i(t_1)[R_{--}(00_i, F; t_1, t_2) - R_{--}(10_i, F; t_1, t_2)] \\
&\quad + F_i(t_2)[R_{++}(11_i, F; t_1, t_2) - R_{++}(10_i, F; t_1, t_2)],
\end{aligned}$$

where $R_s(F; t)$ is the reliability function of system ϕ_s ($s = 1, 2$), $R_{++}(F; t_1, t_2)$ is the joint reliability function of systems ϕ_1, ϕ_2 (the probability that system ϕ_s fails after time t_s , for $s = 1, 2$), and $R_{--}(F; t_1, t_2)$ is the joint unreliability of systems ϕ_1, ϕ_2 (the probability that system ϕ_s fails before time t_s , for $s = 1, 2$). Here, $00_i, 10_i, 11_i$ means component i fails before time t_1 , between times t_1 and t_2 , and after time t_2 , respectively.

4. Signature-based representations for the importance measures

For coherent systems with exchangeable components, Eryilmaz [29] presented a signature-based representation for the Birnbaum importance measure of a component. Here, the related discussions can be generalized to the B-reliability importance measure introduced in Section 2 and the B-TDL importance measure introduced in Section 3. For the two semi-coherent systems ϕ_1 and ϕ_2 in Section 3 with system lifetimes $T^{(1)}$ and $T^{(2)}$ and joint signature $s = (s_{ij}, 1 \leq i, j \leq n)$, let us assume that their component lifetimes $T_1, \dots, T_n$ are i.i.d. from an absolutely continuous distribution function F . Denote $\bar{F} = 1 - F$ and the i th (smallest) order statistic among $T_1, \dots, T_n$ by $T_{i:n}$ ($i = 1, \dots, n$).

From [Definition 2.1](#), we have the B-reliability importance measure defined as $I(i; p) = I_1^{11}(i; p) + I_0^{00}(i; p)$. Then, from the formulas of $I_c^{ab}(i; p)$, we have

$$\begin{aligned}
I_1^{11}(i; p) &= \mathbb{P}\{A_{1,1} | X_i = 1\} - \mathbb{P}\{A_{1,1}\} = \mathbb{P}\{T^{(1)} > t, T^{(2)} > t | T_i > t\} \\
&\quad - \mathbb{P}\{T^{(1)} > t, T^{(2)} > t\}, \\
I_0^{00}(i; p) &= \mathbb{P}\{A_{0,0} | X_i = 0\} - \mathbb{P}\{A_{0,0}\} = \mathbb{P}\{T^{(1)} \leq t, T^{(2)} \leq t | T_i \leq t\} \\
&\quad - \mathbb{P}\{T^{(1)} \leq t, T^{(2)} \leq t\},
\end{aligned}$$

where

$$\begin{aligned}
\mathbb{P}\{T^{(1)} > t, T^{(2)} > t\} &= \sum_{m=0}^{n-1} \left(\sum_{i=m+1}^n \sum_{j=m+1}^n s_{i,j} \right) \binom{n}{m} q^m p^{n-m}, \\
\mathbb{P}\{T^{(1)} \leq t, T^{(2)} \leq t\} &= \sum_{m=1}^n \left(\sum_{i=1}^m \sum_{j=1}^m s_{i,j} \right) \binom{n}{m} q^m p^{n-m},
\end{aligned}$$

with $q = F(t)$, $p = 1 - F(t)$, and

$$\begin{aligned}
\mathbb{P}\{T^{(1)} > t, T^{(2)} > t | T_i > t\} &= \sum_{m=0}^{n-1} \mathbb{P}\{T^{(1)} > t, T^{(2)} > t, S_n(t) = m | T_i > t\} \\
&= \sum_{m=0}^{n-1} \mathbb{P}\{T^{(1)} > t, T^{(2)} > t | S_n(t) = m, T_i > t\} \mathbb{P}\{S_n(t) = m | T_i > t\} \\
&= \sum_{m=0}^{n-1} \frac{r_m^{++i}(n)}{\binom{n-1}{m}} \times \binom{n-1}{m} q^m p^{n-m} p^{-1} = \sum_{m=0}^{n-1} r_m^{++i}(n) q^m p^{n-m-1}, \\
\mathbb{P}\{T^{(1)} \leq t, T^{(2)} \leq t | T_i \leq t\} &= \sum_{m=1}^n \mathbb{P}\{T^{(1)} \leq t, T^{(2)} \leq t, S_n(t) = m | T_i \leq t\} \\
&= \sum_{m=1}^n \mathbb{P}\{T^{(1)} \leq t, T^{(2)} \leq t | S_n(t) = m, T_i \leq t\} \mathbb{P}\{S_n(t) = m | T_i \leq t\} \\
&= \sum_{m=1}^n \frac{r_m^{--i}(n)}{\binom{n-1}{m-1}} \binom{n-1}{m-1} q^m p^{n-m} q^{-1} = \sum_{m=1}^n r_m^{--i}(n) q^{m-1} p^{n-m},
\end{aligned}$$

with $S_n(t)$ being the number of failed components (in systems ϕ_1, ϕ_2) at time t , $r_m^{++i}(n)$ being the number of set A with size m such that component i is not in A and A is not a cut set for both systems ϕ_1, ϕ_2 , and similarly, $r_m^{--i}(n)$ being the number of set A with size m such that component i is in A and A is a cut set for both systems ϕ_1, ϕ_2 . Then, the B-reliability importance measure of component i in systems ϕ_1 and ϕ_2 can be represented as

$$\begin{aligned}
I(i; p) &= \sum_{m=0}^{n-1} \left[r_m^{++i}(n) p^{-1} - \binom{n}{m} \sum_{i=m+1}^n \sum_{j=m+1}^n s_{i,j} \right] q^m p^{n-m} \\
&\quad + \sum_{m=1}^n \left[r_m^{--i}(n) q^{-1} - \binom{n}{m} \sum_{i=1}^m \sum_{j=1}^m s_{i,j} \right] q^m p^{n-m}.
\end{aligned}$$

From [Definition 3.1](#), we have the B-TDL importance measure defined as $I(i; F) = I_{11}^{11}(i; F) + I_{10}^{10}(i; F) + I_{00}^{00}(i; F)$. Then, from the definition of $I_{cd}^{ab}(i; F)$, we have

$$\begin{aligned}
I_{11}^{11}(i; F) &= \mathbb{P}\{A_{1,1} | X_i(t_1) = 1, X_i(t_2) = 1\} - \mathbb{P}\{A_{1,1}\} \\
&= \mathbb{P}\{T^{(1)} > t_1, T^{(2)} > t_2 | T_i > t_2\} - \mathbb{P}\{T^{(1)} > t_1, T^{(2)} > t_2\}, \\
I_{10}^{10}(i; F) &= \mathbb{P}\{A_{1,0} | X_i(t_1) = 1, X_i(t_2) = 0\} - \mathbb{P}\{A_{1,0}\} \\
&= \mathbb{P}\{T^{(1)} > t_1, T^{(2)} \leq t_2 | t_1 < T_i \leq t_2\} - \mathbb{P}\{T^{(1)} > t_1, T^{(2)} \leq t_2\}, \\
I_{00}^{00}(i; F) &= \mathbb{P}\{A_{0,0} | X_i(t_1) = 0, X_i(t_2) = 0\} - \mathbb{P}\{A_{0,0}\} \\
&= \mathbb{P}\{T^{(1)} \leq t_1, T^{(2)} \leq t_2 | T_i \leq t_1\} - \mathbb{P}\{T^{(1)} \leq t_1, T^{(2)} \leq t_2\},
\end{aligned}$$

where

$$\begin{aligned}
\mathbb{P}\{T^{(1)} > t_1, T^{(2)} > t_2\} &= \sum_{m_1=0}^n \sum_{m_2=m_1}^n \left(\sum_{i=m_1+1}^n \sum_{j=m_2+1}^n s_{i,j} \right) \binom{n}{m_1} \\
&\quad \times \binom{n-m_1}{m_2-m_1} P_{m_1, m_2}(t_1, t_2), \\
\mathbb{P}\{T^{(1)} > t_1, T^{(2)} \leq t_2\} &= \sum_{m_1=0}^n \sum_{m_2=m_1}^n \left(\sum_{i=m_1+1}^n \sum_{j=1}^{m_2} s_{i,j} \right) \binom{n}{m_1} \\
&\quad \times \binom{n-m_1}{m_2-m_1} P_{m_1, m_2}(t_1, t_2), \\
\mathbb{P}\{T^{(1)} \leq t_1, T^{(2)} \leq t_2\} &= \sum_{m_1=0}^n \sum_{m_2=m_1}^n \left(\sum_{i=1}^{m_1} \sum_{j=1}^{m_2} s_{i,j} \right) \binom{n}{m_1} \\
&\quad \times \binom{n-m_1}{m_2-m_1} P_{m_1, m_2}(t_1, t_2),
\end{aligned}$$

and

$$\begin{aligned}
& \mathbb{P}\{T^{(1)} > t_1, T^{(2)} > t_2 | T_i > t_2\} \\
&= \sum_{m_1=0}^{n-1} \sum_{m_2=m_1}^{n-1} \mathbb{P}\{T^{(1)} > t_1, T^{(2)} > t_2, S_n(t_1) = m_1, S_n(t_2) = m_2 | T_i > t_2\} \\
&= \sum_{m_1=0}^{n-1} \sum_{m_2=m_1}^{n-1} \mathbb{P}\{T^{(1)} > t_1, T^{(2)} > t_2 | S_n(t_1) = m_1, S_n(t_2) = m_2, T_i > t_2\} \\
&\quad \times \mathbb{P}\{S_n(t_1) = m_1, S_n(t_2) = m_2 | T_i > t_2\} \\
&= \sum_{m_1=0}^{n-1} \sum_{m_2=m_1}^{n-1} \frac{r_{m_1, m_2}^{++i}(n)}{\binom{n-1}{m_1} \binom{n-m_1-1}{m_2-m_1}} \times \binom{n-1}{m_1} \binom{n-m_1-1}{m_2-m_1} P_{m_1, m_2}(t_1, t_2) [\bar{F}(t_2)]^{-1} \\
&= \sum_{m_1=0}^{n-1} \sum_{m_2=m_1}^{n-1} r_{m_1, m_2}^{++i}(n) P_{m_1, m_2}(t_1, t_2) [\bar{F}(t_2)]^{-1}, \\
&\mathbb{P}\{T^{(1)} > t_1, T^{(2)} \leq t_2 | t_1 < T_i \leq t_2\} \\
&= \sum_{m_1=0}^{n-1} \sum_{m_2=m_1+1}^n \mathbb{P}\{T^{(1)} > t_1, T^{(2)} \leq t_2, S_n(t_1) = m_1, S_n(t_2) = m_2 | t_1 < T_i \leq t_2\} \\
&= \sum_{m_1=0}^{n-1} \sum_{m_2=m_1+1}^n \mathbb{P}\{T^{(1)} > t_1, T^{(2)} \leq t_2 | S_n(t_1) = m_1, S_n(t_2) = m_2, t_1 < T_i \leq t_2\} \\
&\quad \times \mathbb{P}\{S_n(t_1) = m_1, S_n(t_2) = m_2 | t_1 < T_i \leq t_2\} \\
&= \sum_{m_1=0}^{n-1} \sum_{m_2=m_1+1}^n \frac{r_{m_1, m_2}^{+-i}(n)}{\binom{n-1}{m_1} \binom{n-m_1-1}{m_2-m_1-1}} \times \binom{n-1}{m_1} \binom{n-m_1-1}{m_2-m_1-1} P_{m_1, m_2}(t_1, t_2) \\
&\quad \times [F(t_2) - F(t_1)]^{-1} \\
&= \sum_{m_1=0}^{n-1} \sum_{m_2=m_1+1}^n r_{m_1, m_2}^{+-i}(n) P_{m_1, m_2}(t_1, t_2) [F(t_2) - F(t_1)]^{-1}, \\
&\mathbb{P}\{T^{(1)} \leq t_1, T^{(2)} \leq t_2 | T_i \leq t_1\} \\
&= \sum_{m_1=1}^n \sum_{m_2=m_1}^n \mathbb{P}\{T^{(1)} \leq t_1, T^{(2)} \leq t_2, S_n(t_1) = m_1, S_n(t_2) = m_2 | T_i \leq t_1\} \\
&= \sum_{m_1=1}^n \sum_{m_2=m_1}^n \mathbb{P}\{T^{(1)} \leq t_1, T^{(2)} \leq t_2 | S_n(t_1) = m_1, S_n(t_2) = m_2, T_i \leq t_1\} \\
&\quad \times \mathbb{P}\{S_n(t_1) = m_1, S_n(t_2) = m_2 | T_i \leq t_1\} \\
&= \sum_{m_1=1}^n \sum_{m_2=m_1}^n \frac{r_{m_1, m_2}^{--i}(n)}{\binom{n-1}{m_1-1} \binom{n-m_1}{m_2-m_1}} \binom{n-1}{m_1-1} \binom{n-m_1}{m_2-m_1} P_{m_1, m_2}(t_1, t_2) [F(t_1)]^{-1} \\
&= \sum_{m_1=1}^n \sum_{m_2=m_1}^n r_{m_1, m_2}^{--i}(n) P_{m_1, m_2}(t_1, t_2) [F(t_1)]^{-1},
\end{aligned}$$

with $S_1(t_1), S_2(t_2)$ being the numbers of failed components (in systems ϕ_1, ϕ_2) at times t_1, t_2 , respectively, $r_{m_1, m_2}^{++i}(n)$ being the number of component set pairs A_1, A_2 with sizes m_1, m_2 such that $i \notin A_1, i \notin A_2, A_1 \subseteq A_2$ and A_s is not a cut set for system ϕ_s ($s = 1, 2$), $r_{m_1, m_2}^{+-i}(n)$ being the number of component set pairs A_1, A_2 with sizes m_1, m_2 such that $i \notin A_1, i \in A_2, A_1 \subseteq A_2$ and A_1 is not a cut set for system ϕ_1 while A_2 is a cut set for system ϕ_2 , $r_{m_1, m_2}^{--i}(n)$ being the number of component set pairs A_1, A_2 with sizes m_1, m_2 such that $i \in A_1 \subseteq A_2$ and A_s is a cut set for system ϕ_s ($s = 1, 2$), and

$$P_{m_1, m_2}(t_1, t_2) = [F(t_1)]^{m_1} [F(t_2) - F(t_1)]^{m_2-m_1} [\bar{F}(t_2)]^{n-m_2}.$$

Then, the B-TDL importance measure of component i in systems ϕ_1 and ϕ_2 can be represented as

$$\begin{aligned}
I(i; F) &= \sum_{m_1=0}^{n-1} \sum_{m_2=m_1}^{n-1} r_{m_1, m_2}^{++i}(n) P_{m_1, m_2}(t_1, t_2) [\bar{F}(t_2)]^{-1} \\
&\quad + \sum_{m_1=0}^{n-1} \sum_{m_2=m_1+1}^n r_{m_1, m_2}^{+-i}(n) P_{m_1, m_2}(t_1, t_2) [F(t_2) - F(t_1)]^{-1} \\
&\quad + \sum_{m_1=1}^n \sum_{m_2=m_1}^n r_{m_1, m_2}^{--i}(n) P_{m_1, m_2}(t_1, t_2) [F(t_1)]^{-1}
\end{aligned}$$

$$- \sum_{m_1=0}^n \sum_{m_2=m_1}^n \left(1 - \sum_{i=1}^{m_1} \sum_{j=m_2+1}^n s_{i,j}\right) \binom{n}{m_1} \binom{n-m_1}{m_2-m_1} P_{m_1, m_2}(t_1, t_2).$$

To describe the structural properties of systems ϕ_1 and ϕ_2 , given the known state of component i , we can readily define conditional signatures for component i as follows:

$$\begin{aligned}
S^{++i}(n) &= \left(S_{m_1, m_2}^{++i}(n), 0 \leq m_1, m_2 \leq n-1 \right), S_{m_1, m_2}^{++i}(n) = \frac{r_{m_1, m_2}^{++i}(n)}{\binom{n-1}{m_1 \wedge m_2} \binom{n-m_1 \wedge m_2-1}{m_2-m_1}}, \\
S^{--i}(n) &= \left(S_{m_1, m_2}^{--i}(n), 1 \leq m_1, m_2 \leq n \right), S_{m_1, m_2}^{--i}(n) = \frac{r_{m_1, m_2}^{--i}(n)}{\binom{n-1}{m_1 \wedge m_2-1} \binom{n-m_1 \wedge m_2}{m_2-m_1}},
\end{aligned}$$

where $S_{m_1, m_2}^{++i}(n)$ is the probability that systems ϕ_1, ϕ_2 are both working with m_1, m_2 failed components not including component i , and similarly, $S_{m_1, m_2}^{--i}(n)$ is the probability that systems ϕ_1, ϕ_2 are both failed with m_1, m_2 failed components including component i , and $r_{m_1, m_2}^{++i}(n), r_{m_1, m_2}^{--i}(n)$ are generalized here for $0 \leq m_1, m_2 \leq n$. If we also define conditional signatures for component i as

$$\begin{aligned}
s^{++i}(n) &= \left(s_{m_1, m_2}^{++i}(n), 1 \leq m_1, m_2 \leq n \right), \\
s^{--i}(n) &= \left(s_{m_1, m_2}^{--i}(n), 1 \leq m_1, m_2 \leq n \right),
\end{aligned}$$

where $s_{m_1, m_2}^{++i}(n), s_{m_1, m_2}^{--i}(n)$ are the probabilities that the m_1 th, m_2 th component failure will cause systems ϕ_1, ϕ_2 to fail, given that component i is working/failed, respectively, then $s^{++i}(n), s^{--i}(n)$ and $S^{++i}(n), S^{--i}(n)$ have the following relationships:

$$\begin{aligned}
S_{m_1, m_2}^{++i}(n) &= \sum_{i=m_1+1}^n \sum_{j=m_2+1}^n s_{i,j}^{++i}(n), & 0 \leq m_1, m_2 \leq n-1, \\
S_{m_1, m_2}^{--i}(n) &= \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} s_{i,j}^{--i}(n), & 1 \leq m_1, m_2 \leq n,
\end{aligned}$$

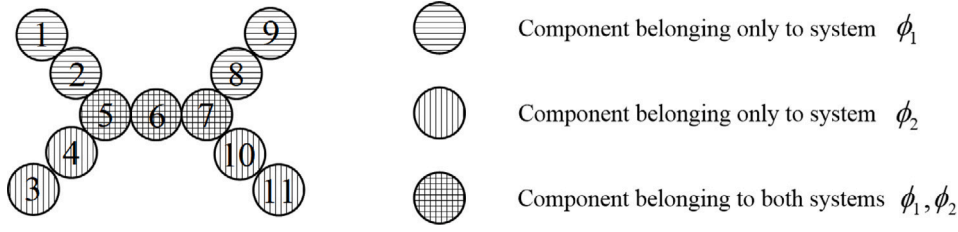
and for $1 \leq m_1, m_2 \leq n$, we also have

$$\begin{aligned}
s_{m_1, m_2}^{++i}(n) &= S_{m_1-1, m_2-1}^{++i}(n) - S_{m_1-1, m_2}^{++i}(n) I_{\{m_2 \neq n\}} - S_{m_1, m_2-1}^{++i}(n) I_{\{m_1 \neq n\}} \\
&\quad + S_{m_1, m_2}^{++i}(n) I_{\{m_1 \neq n, m_2 \neq n\}}, \\
s_{m_1, m_2}^{--i}(n) &= S_{m_1, m_2}^{--i}(n) - S_{m_1-1, m_2}^{--i}(n) I_{\{m_1 \neq 1\}} - S_{m_1, m_2-1}^{--i}(n) I_{\{m_2 \neq 1\}} \\
&\quad + S_{m_1-1, m_2-1}^{--i}(n) I_{\{m_1 \neq 1, m_2 \neq 1\}}.
\end{aligned}$$

5. Illustrative examples

In this section, the theoretical results established in the preceding sections are illustrated with two smart street light systems ϕ_1, ϕ_2 with 11 shared smart street lights. As shown in Fig. 1, smart street lights 1, 2, 8, 9 belong only to smart street light system ϕ_1 , smart street lights 3, 4, 10, 11 belong only to smart street light system ϕ_2 , while smart street lights 5, 6, 7 belong to both systems ϕ_1 and ϕ_2 . Smart street light system ϕ_1 is a linear consecutive-2-out-of-7: F system which fails if and only if there exist at least two consecutive failed ones among its lights 1, 2, 5, 6, 7, 8, 9, and smart street light system ϕ_2 is a system which fails if and only if there exist at least three consecutive failed ones among its lights 3, 4, 5, 6, 7, 10, 11. Note that smart street light systems ϕ_1, ϕ_2 have different parameters since they serve for different types of vehicles on different roads.

Suppose the lifetimes of lights 1, ..., 11 are all i.i.d. from a standard exponential distribution with $F(t) = 1 - e^{-t}, t \geq 0$. Then, the B-reliability importance and the B-TDL importance measures of the 11 shared smart street lights are obtained by Definitions 2.1 and 3.1 based on a finite Markov chain imbedding approach (FMCIA) in Examples 5.1 and 5.2, respectively. The same results can also be obtained by using the signature-based representations in Section 4, and the corresponding discussions for light 6 are presented in Examples 5.3 and 5.4.

Fig. 1. Smart street light systems ϕ_1 and ϕ_2 with shared components.

Example 5.1. To obtain the B-reliability importance measure of smart street lights 1–11, we need to study the joint reliability/unreliability of systems ϕ_1 and ϕ_2 by the finite Markov chain imbedding approach first. By adding lights one by one following the ordering of $1, \dots, 11$, we can define a Markov chain $\{Y(t), t = 1, \dots, 11\}$ with state space

$$S = \{(i_1, i_2) : 0 \leq i_1 \leq 2, 0 \leq i_2 \leq 3\} \\ = \{(0, 0), (0, 1), (0, 2), (0, 3), (1, 0), (1, 1), (1, 2), (1, 3), \\ (2, 0), (2, 1), (2, 2), (2, 3)\},$$

where state (i_1, i_2) means there are exactly consecutive i_s ($s = 1, 2$) lights failed, in the smart street lights just added to system ϕ_s , at time t_s ($i_1 \geq 2$ is condensed to $i_1 = 2$ and $i_2 \geq 3$ is condensed to $i_2 = 3$ for computational efficiency).

For $t_1 \leq t_2$, let us denote $x = 1 - F(t_2)$, $y = F(t_1)$, $z = F(t_2) - F(t_1)$ and $\bar{x} = 1 - x$, $\bar{y} = 1 - y$. Then, for lights 1, 2, 8, 9 belonging only to system ϕ_1 , lights 3, 4, 10, 11 belonging only to system ϕ_2 , and lights 5, 6, 7 belonging to both systems ϕ_1 and ϕ_2 , the transition matrices are (see Box 1), respectively. Now, for lifetimes $T^{(1)}$ and $T^{(2)}$ of systems ϕ_1 and ϕ_2 , the probabilities that $T^{(1)} > t_1$, $T^{(2)} > t_2$ (system ϕ_1 fails after time t_1 and system ϕ_2 fails after time t_2), $T^{(1)} > t_1$, $T^{(2)} \leq t_2$ (system ϕ_1 fails after time t_1 and system ϕ_2 fails before time t_2), $T^{(1)} \leq t_1$, $T^{(2)} > t_2$ (system ϕ_1 fails before time t_1 and system ϕ_2 fails after time t_2), $T^{(1)} \leq t_1$, $T^{(2)} \leq t_2$ (system ϕ_1 fails before time t_1 and system ϕ_2 fails before time t_2) are as follows:

$$R_{++}(F; t_1, t_2) = \mathbb{P}\{T^{(1)} > t_1, T^{(2)} > t_2\} = \pi \left[\prod_{i=1}^{11} A^i(t_1, t_2) \right] u_{11},$$

$$R_{+-}(F; t_1, t_2) = \mathbb{P}\{T^{(1)} > t_1, T^{(2)} \leq t_2\} = \pi \left[\prod_{i=1}^{11} A^i(t_1, t_2) \right] u_{10},$$

$$R_{-+}(F; t_1, t_2) = \mathbb{P}\{T^{(1)} \leq t_1, T^{(2)} > t_2\} = \pi \left[\prod_{i=1}^{11} A^i(t_1, t_2) \right] u_{01},$$

$$R_{--}(F; t_1, t_2) = \mathbb{P}\{T^{(1)} \leq t_1, T^{(2)} \leq t_2\} = \pi \left[\prod_{i=1}^{11} A^i(t_1, t_2) \right] u_{00},$$

where $\pi = (1, 0, \dots, 0)$ and

$$u_{11} = (1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0, 0)^T, u_{10} = (0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0)^T,$$

$$u_{01} = (0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0)^T, u_{00} = (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1)^T.$$

Using Remark 2.2 and the above discussions, the B-reliability importance measure of light i ($i = 1, \dots, 11$) can be given as

$$I(i; x) = \pi \left[\prod_{j=1}^{i-1} A^j(t, t) \right]^{I_{\{i \geq 2\}}} [A^i(0, 0) - A^i(\infty, \infty)] \left[\prod_{j=i+1}^{11} A^j(t, t) \right]^{I_{\{i \leq 10\}}} \\ \times [\bar{x}u_{11} - xu_{00}],$$

where $x = (\underbrace{x, \dots, x}_{11})$, $\bar{x} = 1 - x$, that is,

$$I(1; x) = I(9; x) = 2\bar{x}^8 - 7\bar{x}^7 + 7\bar{x}^6 - 2\bar{x}^4 - \bar{x}^3 + \bar{x}^2,$$

$$I(2; x) = I(8; x) = 2\bar{x}^9 - \bar{x}^8 - 12\bar{x}^7 + 17\bar{x}^6 - \bar{x}^5 - 6\bar{x}^4 - \bar{x}^3 + 2\bar{x}^2,$$

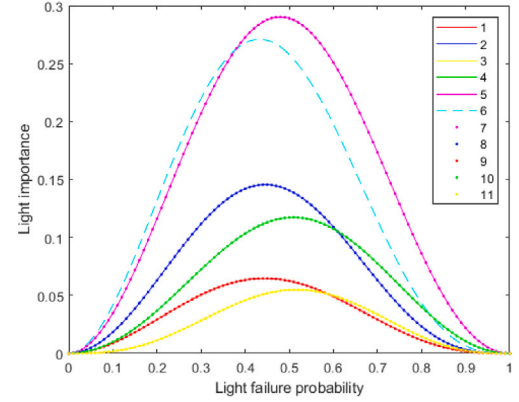


Fig. 2. The B-reliability importance measure for lights 1, ..., 11.

Table 1

Ordering of the B-reliability importance for different $\bar{x}$.

$\bar{x}$	Ordering of B-reliability importance
(0, 0.3786)	6 > 5 > 2 > 4 > 1 > 3
(0.3786, 0.5875)	5 > 6 > 2 > 4 > 1 > 3
(0.5875, 0.6085)	5 > 6 > 2 > 4 > 3 > 1
(0.6085, 0.8198)	5 > 6 > 4 > 2 > 3 > 1
(0.8198, 1)	5 > 4 > 6 > 2 > 3 > 1

$$I(3; x) = I(11; x) = 2\bar{x}^9 - 5\bar{x}^8 - \bar{x}^7 + 11\bar{x}^6 - 7\bar{x}^5 - 2\bar{x}^4 + 2\bar{x}^3,$$

$$I(4; x) = I(10; x) = 2\bar{x}^9 - 5\bar{x}^8 - \bar{x}^7 + 11\bar{x}^6 - 7\bar{x}^5 - \bar{x}^4 + \bar{x}^2,$$

$$I(5; x) = I(7; x) = 2\bar{x}^9 - 3\bar{x}^8 - 8\bar{x}^7 + 17\bar{x}^6 - 5\bar{x}^5 - 2\bar{x}^4 - 5\bar{x}^3 + 4\bar{x}^2,$$

$$I(6; x) = 2\bar{x}^9 - \bar{x}^8 - 11\bar{x}^7 + 12\bar{x}^6 + 5\bar{x}^5 - 4\bar{x}^4 - 8\bar{x}^3 + 5\bar{x}^2.$$

Then, the B-reliability importance measure for lights 1, ..., 11 are plotted in Fig. 2 for different light failure probability $0 \leq \bar{x} \leq 1$. It can be seen from Fig. 2 that lights with symmetric positions (1 and 9, 2 and 8, 3 and 11, 4 and 10, 5 and 7) have the same importance measure in systems ϕ_1, ϕ_2 , and for different $\bar{x}$, and lights 1, ..., 6 can be ordered by their B-reliability importance as in Table 1. In addition, see Appendix A.1 for a plot of $I_c^{ab}(i; x)$, $a, b, c \in \{0, 1\}$, for a better understanding of Remark 2.1.

Remark 5.1. For comparison, the traditional B-reliability importance [2] of light i ($i = 1, \dots, 11$) in the series connection of smart street light systems ϕ_1, ϕ_2 can be given as

$$I(i; x) = \pi \left[\prod_{j=1}^{i-1} A^j(t, t) \right]^{I_{\{i \geq 2\}}} [A^i(0, 0) - A^i(\infty, \infty)] \left[\prod_{j=i+1}^{11} A^j(t, t) \right]^{I_{\{i \leq 10\}}} u_{11},$$

and the same results can be given for parallel connection of them upon replacing u_{11} by $-u_{00}$.

See Fig. 3 for the values for different light failure probability $\bar{x}$. It can be seen from Fig. 3(a) that the ordering of traditional B-reliability importance can often be regarded as $5 > 2 > 6 > 1 > 3 = 4$, which seldom changes with $\bar{x}$. It is interesting to see that lights 3 and 4 have the same importance since the failure of series connection of systems

$$\begin{aligned}
\mathbf{A}^i(t_1, t_2) &= \begin{pmatrix} \bar{y} & 0 & 0 & 0 & y & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \bar{y} & 0 & 0 & 0 & y & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \bar{y} & 0 & 0 & 0 & y & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \bar{y} & 0 & 0 & 0 & y & 0 & 0 & 0 & 0 \\ \bar{y} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y & 0 & 0 & 0 \\ 0 & \bar{y} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y & 0 & 0 \\ 0 & 0 & \bar{y} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y & 0 \\ 0 & 0 & 0 & \bar{y} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y + \bar{y} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y + \bar{y} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y + \bar{y} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y + \bar{y} \end{pmatrix}, i = 1, 2, 8, 9, \\
\mathbf{A}^i(t_1, t_2) &= \begin{pmatrix} x & \bar{x} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ x & 0 & \bar{x} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ x & 0 & 0 & \bar{x} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & x + \bar{x} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & x & \bar{x} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & x & 0 & \bar{x} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & x & 0 & 0 & \bar{x} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & x + \bar{x} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x & \bar{x} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x & 0 & \bar{x} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x & 0 & 0 & \bar{x} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x + \bar{x} \end{pmatrix}, i = 3, 4, 10, 11, \\
\mathbf{A}^i(t_1, t_2) &= \begin{pmatrix} x & z & 0 & 0 & 0 & y & 0 & 0 & 0 & 0 & 0 & 0 \\ x & 0 & z & 0 & 0 & 0 & y & 0 & 0 & 0 & 0 & 0 \\ x & 0 & 0 & z & 0 & 0 & 0 & y & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & x + z & 0 & 0 & 0 & y & 0 & 0 & 0 & 0 \\ x & z & 0 & 0 & 0 & 0 & 0 & 0 & y & 0 & 0 & 0 \\ x & 0 & z & 0 & 0 & 0 & 0 & 0 & 0 & y & 0 & 0 \\ x & 0 & 0 & z & 0 & 0 & 0 & 0 & 0 & 0 & y & 0 \\ 0 & 0 & 0 & x + z & 0 & 0 & 0 & 0 & 0 & 0 & y & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x & y + z & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x & 0 & y + z & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x & 0 & 0 & y + z \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & x + y + z \end{pmatrix}, i = 5, 6, 7,
\end{aligned}$$

Box I.

ϕ_1, ϕ_2 can be caused by light 3 or 4 only by the failure of lights 3, 4, 5. Analogous discussions can be given for Fig. 3(b), and we can see that they are both quite different from Fig. 2.

Example 5.2. Based on the discussions in Example 5.1 and Remark 3.2, the B-TDL importance measure of light i ($i = 1, \dots, 11$) in smart street light systems ϕ_1, ϕ_2 can be given as

$$\begin{aligned}
I(i; F) &= F(t_1)[R_1(1_i, F; t_1) - R_1(0_i, F; t_1)] \\
&\quad + \bar{F}(t_2)[R_2(1_i, F; t_2) - R_2(0_i, F; t_2)] \\
&\quad + \bar{F}(t_1)[R_{--}(00_i, F; t_1, t_2) - R_{--}(10_i, F; t_1, t_2)] \\
&\quad + F(t_2)[R_{++}(11_i, F; t_1, t_2) - R_{++}(10_i, F; t_1, t_2)] \\
&= y[f_{10}(\mathbf{u}_{11} + \mathbf{u}_{10}) - f_{00}(\mathbf{u}_{11} + \mathbf{u}_{10})]
\end{aligned}$$

$$\begin{aligned}
&\quad + x[f_{11}(\mathbf{u}_{11} + \mathbf{u}_{01}) - f_{10}(\mathbf{u}_{11} + \mathbf{u}_{01})] + \bar{y}[f_{00}(\mathbf{u}_{00}) \\
&\quad - f_{10}(\mathbf{u}_{00})] + \bar{x}[f_{11}(\mathbf{u}_{11}) - f_{10}(\mathbf{u}_{11})] \\
&= f_{11}(\mathbf{u}_{11} + x \cdot \mathbf{u}_{01}) + f_{10}(y \cdot \mathbf{u}_{10} - \bar{y} \cdot (\mathbf{u}_{11} + \mathbf{u}_{00}) - x \cdot \mathbf{u}_{01}) \\
&\quad + f_{00}(\bar{y} \cdot \mathbf{u}_{00} - y \cdot (\mathbf{u}_{11} + \mathbf{u}_{10})),
\end{aligned}$$

where $\mathbf{x} = (\underbrace{x, \dots, x}_{11})$, and for $(k, r) \in \{(1, 1), (1, 0), (0, 0)\}$,

$$f_{k,r}(\mathbf{u}) = \pi \left[\prod_{j=1}^{i-1} \mathbf{A}^j(t_1, t_2) \right]^{I_{\{i \geq 2\}}} \mathbf{B}^i(k, r) \left[\prod_{j=i+1}^{11} \mathbf{A}^j(t_1, t_2) \right]^{I_{\{i \leq 10\}}} \mathbf{u},$$

with $\mathbf{B}^i(1, 1) = \mathbf{A}^i(0, 0)$, $\mathbf{B}^i(1, 0) = \mathbf{A}^i(0, \infty)$, $\mathbf{B}^i(0, 0) = \mathbf{A}^i(\infty, \infty)$.

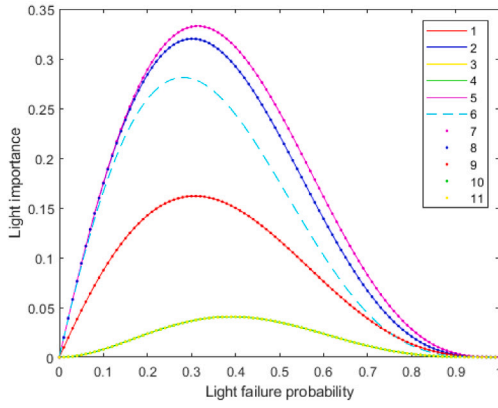
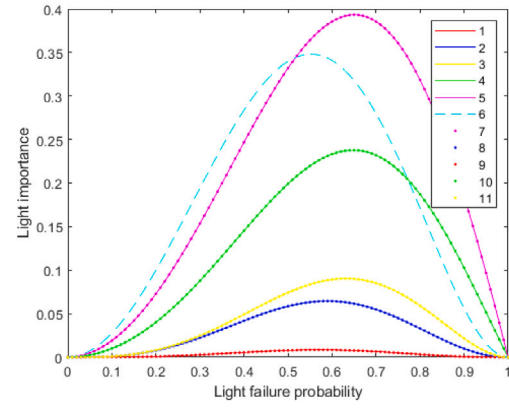
(a) In series connection of systems ϕ_1, ϕ_2 (b) In parallel connection of systems ϕ_1, ϕ_2

Fig. 3. The traditional B-reliability importance for lights 1, ..., 11.

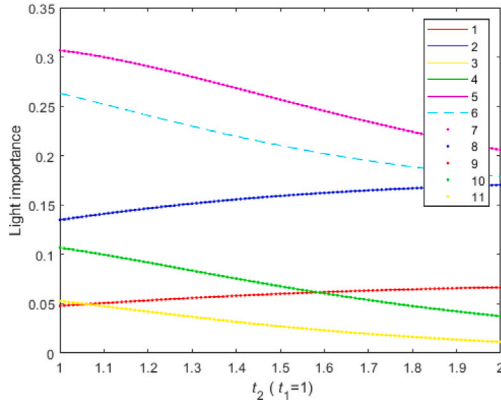


Fig. 4. The B-TDL importance measure for lights 1, ..., 11.

Table 2

Ordering of the B-TDL importance for different t_2 .

t_2	Ordering of B-TDL importance
(1, 1.0599)	$5 > 6 > 2 > 4 > 3 > 1$
(1.0599, 1.5841)	$5 > 6 > 2 > 4 > 1 > 3$
(1.5841, 2)	$5 > 6 > 2 > 1 > 4 > 3$

Then, the B-TDL importance measure for lights 1, ..., 11 have been plotted in Fig. 4 for $1 \leq t_2 \leq 2$, $t_1 = 1$. As can be seen in Fig. 4, the lights with symmetric positions (1 and 9, 2 and 8, 3 and 11, 4 and 10, 5 and 7) have the same importance in smart street light systems ϕ_1, ϕ_2 , and for different t_2 , and lights 1, ..., 6 can be ordered by their B-TDL importance as in Table 2. In addition, see Appendix A.2 for a plot of $I_{cd}^{ab}(i; F)$, $a, b, c, d \in \{0, 1\}$, for a better understanding of Remark 3.1.

Remark 5.2. For comparison, the traditional B-TDL importance [2] of smart street light i ($i = 1, \dots, 11$) in the series connection of systems ϕ_1, ϕ_2 can also be given as

$$I(i; F) = \pi \left[\prod_{j=1}^{i-1} A^j(t, t) \right]^{I_{i \geq 2}} [A^i(0, 0) - A^i(\infty, \infty)] \times \left[\prod_{j=i+1}^{11} A^j(t, t) \right]^{I_{i \leq 10}} u_{11},$$

and the same results can be given for parallel connection of them upon replacing u_{11} by $-u_{00}$.

See Fig. 5 for the values for different time t . Related discussions for Fig. 5(a) are the same as in Remark 5.1 except that the difference in importance of lights 1 and 6 become larger, which leads to a more obvious change in the ordering at time $t \approx 1.4$. Analogous discussions can be given for Fig. 3(b), and we can see that they are both quite different from Fig. 4.

Example 5.3. The B-reliability importance measure for smart street lights 1, ..., 11 can also be obtained by their signature-based representation in Section 4. For example, the B-reliability importance of light 6 can be represented as

$$I(6; p) = \sum_{m=0}^{10} \left[r_m^{++6}(11) p^{-1} - \binom{11}{m} \sum_{i=m+1}^{11} \sum_{j=m+1}^{11} s_{i,j} \right] q^m p^{11-m} + \sum_{m=1}^{11} \left[r_m^{--6}(11) q^{-1} - \binom{11}{m} \sum_{i=1}^m \sum_{j=1}^m s_{i,j} \right] q^m p^{11-m},$$

where the joint signature $s = (s_{i,j}, 1 \leq i, j \leq 11)$ of smart street light systems ϕ_1 and ϕ_2 and number vectors $r^{++6}(11) = (r_0^{++6}(11), \dots, r_{11}^{++6}(11))$, $r^{--6}(11) = (r_0^{--6}(11), \dots, r_{11}^{--6}(11))$ can be given from their definitions as (see the Box II) and $r^{++6}(11) = (1, 10, 41, 88, 104, 64, 16, 0, 0, 0, 0, 0)$, $r^{--6}(11) = (0, 0, 0, 3, 22, 70, 126, 140, 98, 42, 10, 1)$. We then have $I(6; p) = 2q^9 - q^8 - 11q^7 + 12q^6 + 5q^5 - 4q^4 - 8q^3 + 5q^2$, which is the same as the result obtained by the FMCIA in Example 5.1.

Example 5.4. The B-TDL importance measure for smart street lights 1, ..., 11 can also be obtained by their signature-based representation in Section 4. For example, the B-TDL importance measure of light 6 can be represented as

$$I(6; F) = \sum_{m_1=0}^{10} \sum_{m_2=m_1}^{10} r_{m_1, m_2}^{++6}(11) P_{m_1, m_2}(t_1, t_2) [\bar{F}(t_2)]^{-1} + \sum_{m_1=0}^{10} \sum_{m_2=m_1+1}^{11} r_{m_1, m_2}^{+-6}(11) P_{m_1, m_2}(t_1, t_2) [F(t_2) - F(t_1)]^{-1} + \sum_{m_1=1}^{11} \sum_{m_2=m_1}^{11} r_{m_1, m_2}^{--6}(11) P_{m_1, m_2}(t_1, t_2) [F(t_1)]^{-1} - \sum_{m_1=0}^{11} \sum_{m_2=m_1}^{11} \left(1 - \sum_{i=1}^{m_1} \sum_{j=m_2+1}^{11} s_{i,j} \right) \binom{11}{m_1} \binom{11-m_1}{m_2-m_1} \times P_{m_1, m_2}(t_1, t_2),$$

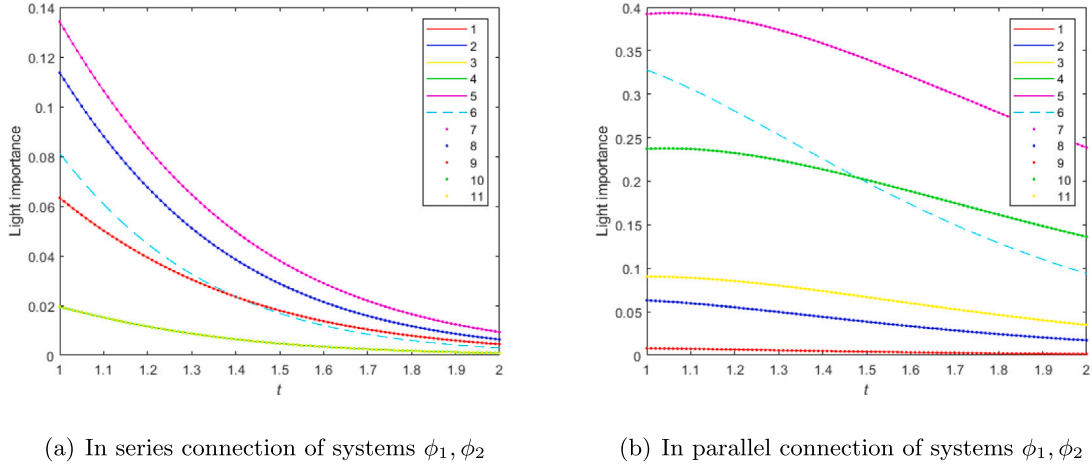


Fig. 5. The traditional B-TDL importance for lights 1, ..., 11.

$s \approx$	0	0	0	0	0	0	0	0	0	0	0
	0	0	0.0081	0.0101	0.0139	0.0179	0.0205	0.0196	0.0140	0.0051	0
	0	0	0.0101	0.0172	0.0238	0.0312	0.0362	0.0350	0.0252	0.0092	0
	0	0	0.0030	0.0242	0.0277	0.0370	0.0431	0.0416	0.0300	0.0115	0
	0	0	0.0035	0.0104	0.0342	0.0343	0.0405	0.0389	0.0279	0.0115	0
	0	0	0.0028	0.0084	0.0169	0.0332	0.0307	0.0294	0.0207	0.0094	0
	0	0	0.0018	0.0053	0.0106	0.0165	0.0208	0.0173	0.0115	0.0058	0
	0	0	0.0008	0.0025	0.0049	0.0076	0.0084	0.0061	0.0040	0.0020	0
	0	0	0.0002	0.0006	0.0013	0.0019	0.0019	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0

Box II.

where the joint signature $s = (s_{i,j}, 1 \leq i, j \leq 11)$ of smart street light systems ϕ_1 and ϕ_2 is the same as given above while number matrices $r^{++6}(11) = (r_{i,j}^{++6}(11), 0 \leq i, j \leq 11)$, $r^{+-6}(11) = (r_{i,j}^{+-6}(11), 0 \leq i, j \leq 11)$, $r^{--6}(11) = (r_{i,j}^{--6}(11), 0 \leq i, j \leq 11)$ are given by (see Box III) which leads to the same result as in Fig. 3 and also leads to the following conditional signatures (see Box IV).

6. Concluding remarks

In this work, we have proposed two new importance measures (B-reliability importance and B-TDL importance measures) for two semi-coherent systems having shared components for describing the importance of the shared components in both systems. Compared with existing importance measures, the proposed measures evaluate the importance of shared components in the systems that share them instead of the importance of components in single systems, and this aspect has not been studied before. Signature-based representations are also provided for the computation of the proposed measures. The established results have been applied to consecutive- k systems with shared components for the purpose of demonstration. Possible applications of this work include, but are not limited to, telecommunication systems, oil pipeline systems, sensing systems, smart street light systems

and unmanned driving systems. Moreover, the present work can be generalized to the following situations:

- (1) More different types of importance measures can be studied and compared for components shared by two or more semi-coherent systems;
- (2) Systems considered in this work are binary-state ones with binary-state components, and related discussions can be generalized to multi-state systems with shared multi-state components;
- (3) Components in this work are assumed to be *i.i.d.*, and the effect of dependence among the components can be considered to see how corresponding importance measures change;
- (4) The importance measures proposed here evaluate the importance of single shared component in two semi-coherent systems, and related results can be generalized to pairs/groups of shared components;
- (5) Orderings of importance measures can be studied for consecutive k -type systems with shared components with potential applications to the problems of optimal system design and optimal component allocation as in [34–36].

We are currently working on all these problems and hope to report the findings in a future paper.

$$\begin{aligned}
 r^{++6}_{(11)} &= \begin{pmatrix} 1 & 10 & 45 & 118 & 196 & 210 & 141 & 54 & 9 & 0 & 0 & 0 \\ 10 & 10 & 90 & 354 & 784 & 1050 & 846 & 378 & 72 & 0 & 0 & 0 \\ 41 & 82 & 41 & 322 & 1066 & 1894 & 1895 & 1008 & 222 & 0 & 0 & 0 \\ 90 & 270 & 270 & 88 & 578 & 1522 & 1998 & 1302 & 336 & 0 & 0 & 0 \\ 116 & 464 & 696 & 452 & 104 & 536 & 1025 & 858 & 265 & 0 & 0 & 0 \\ 90 & 450 & 900 & 874 & 398 & 64 & 232 & 272 & 104 & 0 & 0 & 0 \\ 41 & 246 & 615 & 794 & 537 & 168 & 16 & 32 & 16 & 0 & 0 & 0 \\ 10 & 70 & 210 & 338 & 302 & 138 & 24 & 0 & 0 & 0 & 0 & 0 \\ 1 & 8 & 28 & 54 & 60 & 36 & 9 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \\
 r^{+-6}_{(11)} &= \begin{pmatrix} 0 & 0 & 0 & 3 & 22 & 70 & 126 & 140 & 98 & 42 & 10 & 1 \\ 0 & 0 & 0 & 6 & 66 & 280 & 630 & 840 & 686 & 336 & 90 & 10 \\ 0 & 0 & 0 & 3 & 62 & 388 & 1156 & 1920 & 1878 & 1072 & 328 & 41 \\ 0 & 0 & 0 & 0 & 18 & 220 & 970 & 2130 & 2590 & 1768 & 630 & 90 \\ 0 & 0 & 0 & 0 & 0 & 43 & 372 & 1206 & 1932 & 1635 & 696 & 116 \\ 0 & 0 & 0 & 0 & 0 & 0 & 52 & 328 & 770 & 854 & 450 & 90 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 33 & 148 & 238 & 164 & 41 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 10 & 30 & 30 & 10 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \\
 r^{--6}_{(11)} &= \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 30 & 98 & 182 & 210 & 154 & 70 & 18 & 2 \\ 0 & 0 & 0 & 3 & 52 & 272 & 704 & 1050 & 948 & 512 & 152 & 19 \\ 0 & 0 & 0 & 24 & 22 & 242 & 976 & 2000 & 2310 & 1522 & 532 & 76 \\ 0 & 0 & 0 & 84 & 154 & 70 & 580 & 1827 & 2870 & 2398 & 1014 & 169 \\ 0 & 0 & 0 & 168 & 462 & 420 & 126 & 808 & 1934 & 2180 & 1160 & 232 \\ 0 & 0 & 0 & 210 & 770 & 1050 & 630 & 140 & 678 & 1156 & 824 & 206 \\ 0 & 0 & 0 & 168 & 770 & 1400 & 1260 & 560 & 98 & 336 & 360 & 120 \\ 0 & 0 & 0 & 84 & 462 & 1050 & 1260 & 840 & 294 & 42 & 90 & 45 \\ 0 & 0 & 0 & 24 & 154 & 420 & 630 & 560 & 294 & 84 & 10 & 10 \\ 0 & 0 & 0 & 3 & 22 & 70 & 126 & 140 & 98 & 42 & 10 & 1 \end{pmatrix},
 \end{aligned}$$

Box III.

CRedit authorship contribution statement

He Yi: Writing – original draft, Validation, Software, Methodology, Conceptualization. **Narayanaswamy Balakrishnan:** Writing – original draft, Validation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

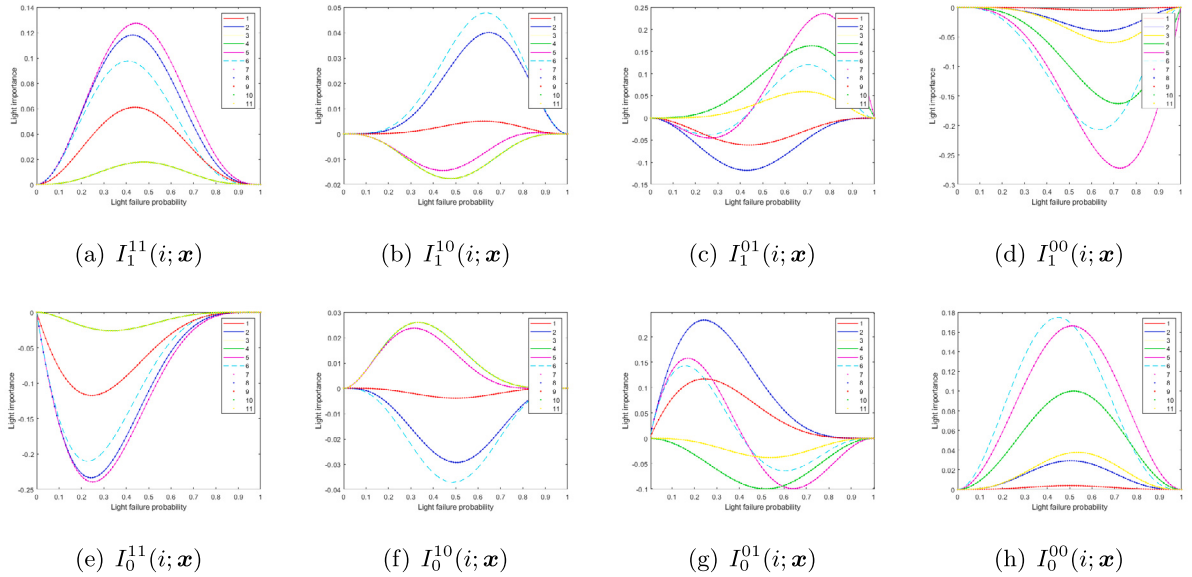
No data was used for the research described in the article.

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$$\begin{aligned}
 s^{++i}(11) &\approx \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.0016 & 0.0056 & 0.0119 & 0.0198 & 0.0262 & 0.0238 & 0 & 0 \\ 0 & 0 & 0 & 0.0032 & 0.0103 & 0.0217 & 0.0359 & 0.0471 & 0.0429 & 0 & 0 \\ 0 & 0 & 0.0024 & 0.0024 & 0.0143 & 0.0283 & 0.0446 & 0.0565 & 0.0492 & 0 & 0 \\ 0 & 0 & 0.0040 & 0.0119 & 0.0079 & 0.0302 & 0.0449 & 0.0535 & 0.0429 & 0 & 0 \\ 0 & 0 & 0.0041 & 0.0124 & 0.0248 & 0.0127 & 0.0381 & 0.0413 & 0.0286 & 0 & 0 \\ 0 & 0 & 0.0033 & 0.0100 & 0.0200 & 0.0310 & 0.0095 & 0.0254 & 0.0127 & 0 & 0 \\ 0 & 0 & 0.0021 & 0.0062 & 0.0124 & 0.0190 & 0.0214 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.0008 & 0.0024 & 0.0048 & 0.0071 & 0.0071 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \\
 s^{--i}(11) &\approx \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.0444 & 0.0389 & 0.0333 & 0.0278 & 0.0222 & 0.0167 & 0.0111 & 0.0056 & 0 \\ 0 & 0 & 0.0222 & 0.0389 & 0.0381 & 0.0357 & 0.0317 & 0.0262 & 0.0190 & 0.0103 & 0 \\ 0 & 0 & 0 & 0.0389 & 0.0333 & 0.0357 & 0.0349 & 0.0310 & 0.0238 & 0.0135 & 0 \\ 0 & 0 & 0 & 0 & 0.0452 & 0.0278 & 0.0308 & 0.0295 & 0.0240 & 0.0141 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.0397 & 0.0216 & 0.0229 & 0.0197 & 0.0121 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.0254 & 0.0143 & 0.0127 & 0.0079 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.0095 & 0.0063 & 0.0032 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.
 \end{aligned}$$

Box IV.

Fig. 6. $I_c^{ab}(i; \mathbf{x})$, $a, b, c \in \{0, 1\}$, in Example 5.1.

Appendix

A.1. Figure of $I_c^{ab}(i; \mathbf{x})$, $a, b, c \in \{0, 1\}$, in Example 5.1

See Fig. 6.

A.2. Figure of $I_{cd}^{ab}(i; F)$, $a, b, c, d \in \{0, 1\}$, in Example 5.2

See Fig. 7.

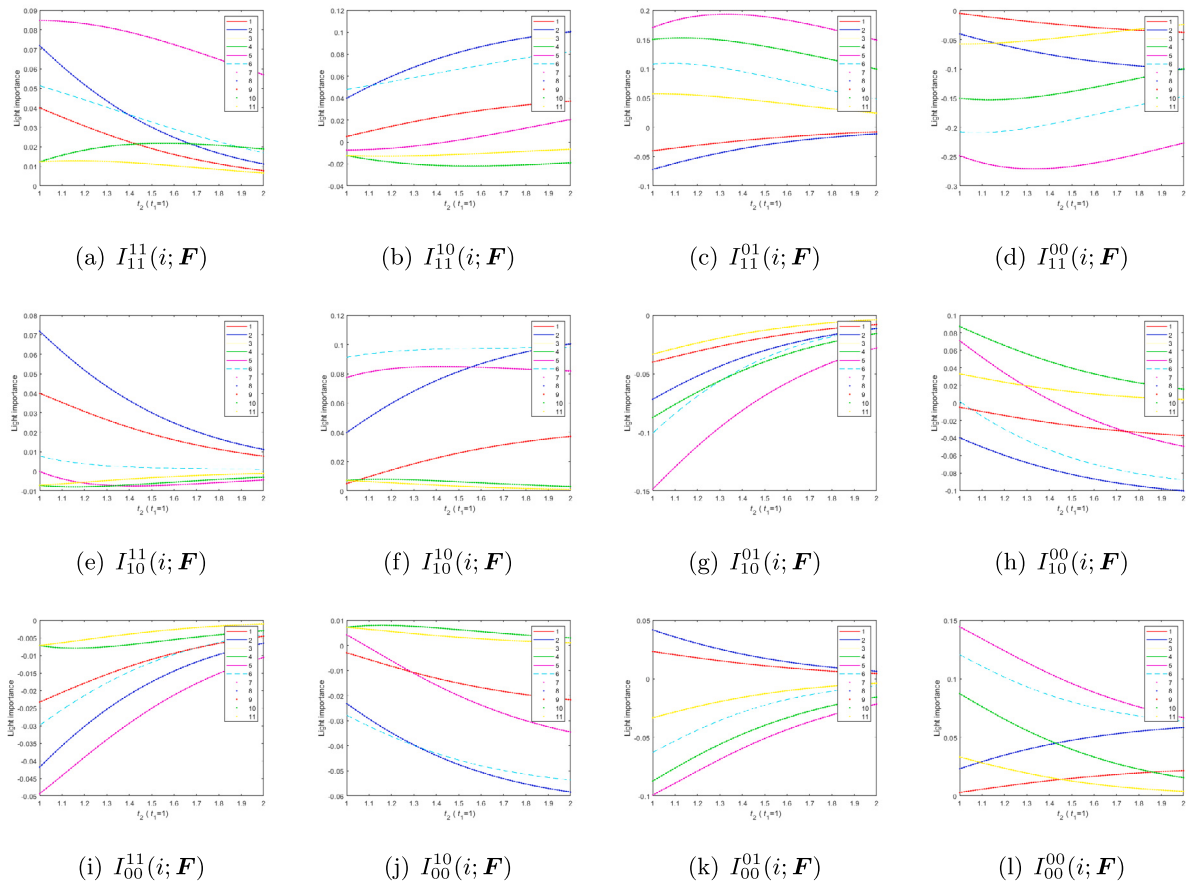


Fig. 7. $I_{cd}^{ab}(i; F)$, $a, b, c, d \in \{0, 1\}$, in Example 5.2.

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